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Managing database offsets with time series

Abstract

Systems and methods of improving the functioning of a streaming platform system by managing database change stream offsets using a time series database are disclosed. In some example embodiments, a computer system retrieves an offset value from a plurality of offset values stored in a time series database, with the plurality of offset values being indexed in the time series database in time order, and the retrieved offset value being retrieved using a time parameter, and then the computer system transmits a data request to a stream-processing platform, with the data request comprising the retrieved offset value, and the data request being operable to retrieve a data record stored in association with the retrieved offset value in a storage layer of the stream-processing platform using the offset value.

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Background/Summary

RELATED APPLICATIONS (1) This application is a continuation of and claims priority to U.S. patent application Ser. No. 17/536,705 filed Nov. 29, 2021, which claims priority to U.S. patent application Ser. No. 16/393,404, filed Apr. 24, 2019, entitled “Managing Database Offsets With Time Series”, the entire disclosures of which are hereby incorporated by reference herein in their entirety.

TECHNICAL FIELD

(1) Embodiments of the present disclosure relate generally to the technical field of an electrical computer system architecture and, more particularly, but not by way of limitation, to systems and methods of improving the functioning of a streaming platform system by managing database offsets using a time series database.

BACKGROUND

(2) A streaming platform system processes incoming streams of data records of an online service. For example, a stream of data records may comprise data change events comprising changes to content of an online service, which may be processed by the streaming platform system as those changes are submitted by producers of the content. However, streaming platform systems suffer from technical problems associated with consumers (e.g., consuming processes) of the streaming platforms systems accessing all of the data change events, such as when a consumer attempts to start, or restart, an application at a particular point in a data stream (e.g., attempting to back up or replay the data stream from an earlier point in time in order to recover from a system failure). Current solutions for a consumer to access starting an application at a particular point in a data stream on a streaming platform system rely on distributed file-based storage or other inefficient mechanisms that require extensive computer processing activity, thereby resulting in excessive consumption of electronic resources (e.g., processing power, memory, network bandwidth). Additionally, current solutions require specific versions of an application program interface (API) to access the offsets that are used to identify the particular point in the data stream at which to start the application, thereby limiting access and use of the streaming platform system. Furthermore, current solutions do not efficiently enable consumers to accurately identify the appropriate point in the data stream at which to start the application or otherwise access data. Other technical problems can arise as well.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Some embodiments of the present disclosure are illustrated by way of example and not

limitation in the figures of the accompanying drawings, in which like reference numbers indicate similar elements.

(2) FIG. 1 is a block diagram illustrating a networked system, in accordance with some example embodiments.

(3) FIG. 2 illustrates a streaming platform system, in accordance with some example embodiments.

(4) FIG. 3 illustrates a time series database, in accordance with some example embodiments.

(5) FIG. 4 illustrates a distributed data store for log files, in accordance with some example embodiments.

(6) FIG. 5 is a flowchart illustrating a method of improving the functioning of a streaming platform system by managing database change stream offsets using a time series database, in accordance with some example embodiments.

(7) FIG. 6 is a flowchart illustrating another method of improving the functioning of a streaming platform system by managing database change stream offsets using a time series database, in accordance with some example embodiments.

(8) FIG. 7 is a block diagram illustrating various components of a network-based publication system, in accordance with some example embodiments.

(9) FIG. 8 is a block diagram illustrating various tables that can be maintained within a database, in accordance with some example embodiments.

(10) FIG. 9 is a block diagram illustrating a representative software architecture, in accordance with some example embodiments.

(11) FIG. 10 is a diagrammatic representation of a machine in the form of a computer system within which a set of instructions may be executed for causing the machine to perform any one or more of the methodologies discussed herein, according to an example embodiment.

DETAILED DESCRIPTION

(12) The description that follows includes illustrative systems, methods, techniques, instruction sequences, and computing machine program products that embody illustrative embodiments. In the following description, for purposes of explanation, numerous specific details are set forth in order to provide an understanding of various embodiments of the inventive subject matter. It will be evident, however, to those skilled in the art that embodiments of the inventive subject matter can be practiced without these specific details. In general, well-known instruction instances, protocols, structures, and techniques have not been shown in detail.

(13) The present disclosure provides technical solutions for improving the functioning of a streaming platform system, or any other type of computer system, by managing database offsets using a time series database. In some example embodiments, a computer system having a memory and at least one hardware processor uses a time series database to store offsets for a streaming platform system so that a data stream of the streaming platform system may be accurately, effectively, and efficiently repositioned to a particular time (e.g., to just before a system failure) by using a time parameter corresponding to that particular time. In some example embodiments, the time parameter is used by the computer system to retrieve an offset value from the time series database, and the retrieved offset value is then used to reposition the data stream of the streaming platform system to a point corresponding with the offset value or to otherwise retrieve a data record corresponding to the particular time of the time parameter.

(14) The implementation of the features disclosed herein involves a non-generic, unconventional, and non-routine operation or combination of operations. By applying one or more of the solutions disclosed herein, some technical effects of the system and method of the present disclosure are to improve the accuracy, effectiveness, and efficiency of accessing specific points in a data stream of a streaming platform computer system. The functional operations and features disclosed herein facilitate recovery of the streaming platform computer system from network or system failure and reduce data query errors. For example, because times may be used to query for data, the exact data needed for analysis may be retrieved—in contrast, if offsets are used, more of the stream may be

required because of uncertainty of the stream data required or an incorrect amount of the stream data may be retrieved. In this way, providing a mapping between offsets and time allows for more data accuracy in the queries as well as improved network bandwidth. Further still, because the exact data required for analysis is queried, it reduces the processor cycles required for analysis since processor cycles are not expended on analysis of unnecessary data retrieved by offsets. Additionally, the use of a time series database to manage offsets, as disclosed herein, reduces dependency on a specific version of an API to identify and access a particular point in the data stream, thereby improving the flexibility and accessibility of the streaming platform computer system. As a result, the functioning of the computer system is improved because it is more resilient to component changes. Other improvements to the functioning of a computer or machine are also apparent from this disclosure.

(15) In some example embodiments, operations are performed by a computer system or other machine having a memory and at least one hardware processor, with the operations comprising: retrieving an offset value from a plurality of offset values stored in a time series database, the plurality of offset values being indexed in the time series database in time order, and the retrieved offset value being retrieved using a time parameter; and transmitting a data request to a stream-processing platform, the data request comprising the retrieved offset value, and the data request being operable to retrieve a data record stored in association with the retrieved offset value in a storage layer of the stream-processing platform using the offset value. In some example embodiments, the data request is further operable to reposition a stream of data published from the stream-processing platform to an application at a position corresponding to the offset value.

(16) In some example embodiments, the operations further comprises: prior to the retrieving of the offset value, receiving a series of data records; for each one of the data records in the series of data records, storing the one of the data records in the storage layer of the stream-processing platform in association with a corresponding one of the plurality of offset values; and for each one of the plurality of offset values, storing the one of the plurality of offset values in the time series database.

(17) In some example embodiments, the time parameter comprises a single point in time. In some example embodiments, the time parameter comprises a time range having a start time and an end time. However, other indications of time may be used as the time parameter as well.

(18) In some example embodiments, the data record comprises change data indicating at least one change to content of an online site. However, the data record may comprise other types of data as well.

(19) In some example embodiments, the retrieving of the offset value is performed in response to an interruption of the stream-processing platform publishing a stream of data to an application. However, the retrieving of the offset value may be triggered in other ways as well.

(20) The methods or embodiments disclosed herein can be implemented as a computer system having one or more modules (e.g., hardware modules or software modules). Such modules can be executed by one or more hardware processors of the computer system. The methods or embodiments disclosed herein can be embodied as instructions stored on a machine-readable medium that, when executed by one or more processors, cause the one or more processors to perform the instructions.

(21) With reference to FIG. 1, an example embodiment of a high-level client-server-based network architecture **100** is shown. A networked system **102**, in the example forms of a network-based marketplace or payment system, provides server-side functionality via a network **104** (e.g., the Internet or wide area network (WAN)) to one or more client devices **110**. FIG. 1 illustrates, for example, a web client **112** (e.g., a browser, such as the Internet Explorer® browser developed by Microsoft Corporation of Redmond, Washington State), an application **114**, and a programmatic client **116** executing on client device **110**.

(22) The client device **110** may comprise, but is not limited to, a mobile phone, desktop computer, laptop, portable digital assistants (PDAs), smart phones, tablets, ultra books, netbooks, laptops,

multi-processor systems, microprocessor-based or programmable consumer electronics, game consoles, set-top boxes, or any other communication device that a user may utilize to access the networked system **102**. In some embodiments, the client device **110** may comprise a display module (not shown) to display information (e.g., in the form of user interfaces). In further embodiments, the client device **110** may comprise one or more of a touch screens, accelerometers, gyroscopes, cameras, microphones, global positioning system (GPS) devices, and so forth. The client device **110** may be a device of a user that is used to perform a transaction involving digital items within the networked system **102**. In one embodiment, the networked system **102** is a network-based marketplace that responds to requests for product listings, publishes publications comprising item listings of products available on the network-based marketplace, and manages payments for these marketplace transactions. One or more users **106** may be a person, a machine, or other means of interacting with client device **110**. In embodiments, the user **106** is not part of the network architecture **100**, but may interact with the network architecture **100** via client device **110** or another means. For example, one or more portions of network **104** may be an ad hoc network, an intranet, an extranet, a virtual private network (VPN), a local area network (LAN), a wireless LAN (WLAN), a wide area network (WAN), a wireless WAN (WWAN), a metropolitan area network (MAN), a portion of the Internet, a portion of the Public Switched Telephone Network (PSTN), a cellular telephone network, a wireless network, a WiFi network, a WiMax network, another type of network, or a combination of two or more such networks.

(23) Each client device **110** may include one or more applications (also referred to as “apps”) such as, but not limited to, a web browser, messaging application, electronic mail (email) application, an e-commerce site application (also referred to as a marketplace application), and the like. In some embodiments, if the e-commerce site application is included in a given client device **110**, then this application is configured to locally provide the user interface and at least some of the functionalities with the application configured to communicate with the networked system **102**, on an as needed basis, for data and/or processing capabilities not locally available (e.g., access to a database of items available for sale, to authenticate a user, to verify a method of payment, etc.). Conversely if the e-commerce site application is not included in the client device **110**, the client device **110** may use its web browser to access the e-commerce site (or a variant thereof) hosted on the networked system **102**.

(24) One or more users **106** may be a person, a machine, or other means of interacting with the client device **110**. In example embodiments, the user **106** is not part of the network architecture **100**, but may interact with the network architecture **100** via the client device **110** or other means. For instance, the user **106** provides input (e.g., touch screen input or alphanumeric input) to the client device **110** and the input is communicated to the networked system **102** via the network **104**. In this instance, the networked system **102**, in response to receiving the input from the user **106**, communicates information to the client device **110** via the network **104** to be presented to the user **106**. In this way, the user **106** can interact with the networked system **102** using the client device **110**.

(25) An application program interface (API) server **120** and a web server **122** are coupled to, and provide programmatic and web interfaces respectively to, one or more application servers **140**. The application servers **140** may host one or more publication systems **142**, payment systems **144**, and a streaming platform system **150**, each of which may comprise one or more modules or applications and each of which may be embodied as hardware, software, firmware, or any combination thereof. The application servers **140** are, in turn, shown to be coupled to one or more database servers **124** that facilitate access to one or more information storage repositories or database(s) **126**. In an example embodiment, the databases **126** are storage devices that store information to be posted (e.g., publications or listings) to the publication system **142**. The databases **126** may also store digital item information in accordance with example embodiments.

(26) Additionally, a third party application **132**, executing on third party server(s) **130**, is shown as

having programmatic access to the networked system **102** via the programmatic interface provided by the API server **120**. For example, the third party application **132**, utilizing information retrieved from the networked system **102**, supports one or more features or functions on a website hosted by the third party. The third party website, for example, provides one or more promotional, marketplace, or payment functions that are supported by the relevant applications of the networked system **102**.

(27) The publication systems **142** provides a number of publication functions and services to users **106** that access the networked system **102**. The payment systems **144** likewise provide a number of functions to perform or facilitate payments and transactions. While the publication system **142** and payment system **144** are shown in FIG. **1** to both form part of the networked system **102**, it will be appreciated that, in alternative embodiments, each system **142** and **144** may form part of a payment service that is separate and distinct from the networked system **102**. In some embodiments, the payment systems **144** may form part of the publication system **142**.

(28) The streaming platform system **150** provides functionality operable to perform various stream processing operations, as will be discussed in further detail below. The streaming platform system **150** may access the data from the databases **126**, the third party servers **130**, the publication system **142**, and other sources. In some example embodiments, the streaming platform system **150** may analyze the data to perform stream processing operations. In some example embodiments, the streaming platform system **150** communicates with the publication systems **142** (e.g., accessing item listings) and payment system **144**. In an alternative embodiment, the streaming platform system **150** is a part of the publication system **142**.

(29) Further, while the client-server-based network architecture **100** shown in FIG. **1** employs a client-server architecture, the present inventive subject matter is of course not limited to such an architecture, and could equally well find application in a distributed, or peer-to-peer, architecture system, for example. The publication system **142**, payment system **144**, and streaming platform system **150** can also be implemented as standalone software programs, which do not necessarily have networking capabilities.

(30) The web client **112** may access the various publication and payment systems **142** and **144** via the web interface supported by the web server **122**. Similarly, the programmatic client **116** accesses the various services and functions provided by the publication and payment systems **142** and **144** via the programmatic interface provided by the API server **120**. The programmatic client **116** may, for example, be a seller application (e.g., the Turbo Lister application developed by eBay® Inc., of San Jose, California) to enable sellers to author and manage listings on the networked system **102** in an off-line manner, and to perform batch-mode communications between the programmatic client **116** and the networked system **102**.

(31) Additionally, a third party application(s) **132**, executing on a third party server(s) **130**, is shown as having programmatic access to the networked system **102** via the programmatic interface provided by the API server **120**. For example, the third party application **132**, utilizing information retrieved from the networked system **102**, may support one or more features or functions on a website hosted by the third party. The third party website may, for example, provide one or more promotional, marketplace, or payment functions that are supported by the relevant applications of the networked system **102**.

(32) FIG. **2** illustrates the streaming platform system **150**, in accordance with some example embodiments. The streaming platform system **150** is configured to perform the operations and implement the features disclosed herein. In some example embodiments, the streaming platform system **150** comprises any combination of one or more of a database **210**, a distributed data store **220** for log files **222**, a change stream processing application **230**, a time series database **240**, an application starter **250**, and a data repository **260**. The database **210**, the distributed data store **220**, the change stream processing application **230**, the time series database **240**, the application starter **250**, and the data repository **260** are communicatively coupled to each other, such as via a

communication network (e.g., network **104** in FIG. **1**). In some example embodiments, the database **210**, the distributed data store **220**, the change stream processing application **230**, the time series database **240**, the application starter **250**, and the data repository **260** reside on a single machine having a memory and at least one hardware processor. In some example embodiments, one or more of the database **210**, the distributed data store **220**, the change stream processing application **230**, the time series database **240**, the application starter **250**, and the data repository **260** reside on different machines. The database **210**, the distributed data store **220**, the time series database **240**, and the data repository **260**, or a portion thereof, can be incorporated into the database(s) **126** of FIG. **1**.

(33) In some example embodiments of the streaming platform system **150** is configured to process streams of data records as they occur, storing them in a fault-tolerant and durable way, and publishing the streams of data records for use by consumers. A stream refers to constant incoming flow of messages (e.g., messages of a similar type or category). For example, a stream of data records may comprise all the updates to a database, all the logs produced by a service, or any other type of event data. The term “streaming” may refer to processing streams of incoming data in real-time.

(34) In some example embodiments, the database **210** stores any changes to data of an online service (e.g., content of a website), such as insert, update, or delete data activity, as the changes commit to one or more tables of the online service. A commit is the final step in the successful completion of a previously started database change as part of handling a transaction in a computing system, making one or more tentative changes permanent. In some example embodiments, the database **210** comprises a change data capture (CDC) sink **212** configured to detect, capture, receive, or otherwise determine the committed data changes, and then store indications of the committed data changes in log files **222** (e.g., log files **222-1**, . . . , **222-N**) of the distributed data store **220**. In this way, as inserts, updates, and deletes are applied to tracked source tables of the database **212**, entries that describe those changes are added to a commit log in the distributed data store **220**. In some example embodiments, the CDC sink **212** converts the change data into a data structure having a change event format comprising the delta of the change event, which is then stored as a data record in a log file **222** in the distributed data store **220**.

(35) In some example embodiments, the change stream processing application **230** is configured to consume the data records stored in the log files **222** of the distributed data store **220**, using the consumed data records as source data **232**, which is processed via one or more transformation operations **234** to convert the data records from one format into another format. The processed data records may then be written to a sink **236**. In some example embodiments, the sink **236** is configured to store the processed data records in the data repository **260** for use by one or more consumers of the streaming platform system **150**, such as applications that use the streaming data of the streaming platform system **150** for display to end users on client devices.

(36) In some example embodiments, the components of the streaming platform system **150** are run as a cluster on one or more servers called brokers that may span multiple datacenters. The cluster may store streams of data records in categories called topics. A topic is a category or feed name to which data records are published. The streams of data records may be received as messages from producers, and the data may be partitioned into different partitions with different topics that may be distributed across cluster nodes. Each partition is an ordered, immutable sequence of data records that is continually appended to (e.g., a structured commit log). The data records in the partitions may each be assigned a sequential identification number called the offset that uniquely identifies each data record within the partition. Within a partition, data records may be ordered, indexed, and stored by their offsets (e.g., the position of a data record within a partition).

(37) Other processes called consumers can read data records from partitions. Consumers may label themselves with a consumer group name, and each record published to a topic may be delivered to

one consumer instance within each subscribing consumer group. Consumer instances can be in separate processes or on separate machines. A producer API may enable an application to publish streams of data records via the streaming platform system **150**, while a consumer API may enable an application to subscribe to topics and process streams of data records. In some example embodiments, the streaming platform system **150** durably persists all published records, whether or not they have been consumed, using a configurable retention period. The partitions of the log file may be distributed over the servers in the cluster with each server handling data and requests for a share of the partitions, and each partition may be replicated across a configurable number of servers for fault tolerance.

(38) In some example embodiments, the change stream processing application **230** comprises a stream processor that is configured to take continual streams of data records from input topics, perform some processing on this input, and produce continual streams of data to output topics. For example, a retail application may take in input streams of sales and shipments and output a stream of reorders and price adjustments computed off this data. In some example embodiments, raw input data is consumed from topics and then aggregated, enriched, or otherwise transformed into new topics for further consumption or follow-up processing. For example, a processing pipeline for recommending content may crawl online content from data feeds and publish the content to a particular topic, further processing may normalize or deduplicate this content and publish the cleansed content to a new topic, and a final processing stage may attempt to recommend this content to users. Such processing pipelines may create graphs of real-time data flows based on the individual topics.

(39) Referring back to FIG. 2, in some example embodiments, the sink **236** is configured to, in response to or otherwise based on the storing of a processed data record in the data repository **260**, store the corresponding offset value of the processed data record in the time series database **240** in association with a corresponding timestamp. FIG. 3 illustrates the time series database **240**, in accordance with some example embodiments. In the example embodiment of FIG. 3, multiple offset values (e.g., OFFSET-1, OFFSET-2, . . . , OFFSET-N) are each stored in association with a corresponding timestamp (e.g., TIMESTAMP-1, TIMESTAMP-2, . . . , TIMESTAMP-N). In some example embodiments, each offset value is also stored in association with one or more other attributes, such as an identification of a corresponding datacenter for the data record to which the offset value corresponds (e.g., DATACENTER-1, DATACENTER-2, DATACENTER-3, . . .) and an identification of a corresponding topic for the data record to which the offset value corresponds (e.g., TOPIC-1, TOPIC-2, TOPIC-3, . . .). In some example embodiments, each offset value belongs to a particular topic and partition along with other key informational labels. A topic may have many partitions and, therefore, can have many offset values as partitions for the topic for each collection of metrics. Notably, the timestamps along with the offsets may be stored during data production—synchronously or asynchronously with the data production. Other attributes are also within the scope of the present disclosure. One example of a schema for the time series database **240** is provided below with example values: name: “partitionOffsets” namespace: “nudata” subsystem: “gsifeeder” labels and values feeder: “gsi” region: “NA” datacenter: “SLC” keyspace: “ItemListings” topic: “item.t2” partition: 1 offset: 1055 timestamp: 04-17-2019: 13:21:09 clusterName: xxx componentId: “3081” instance: “10.147.171.87:9500” job: “gsi-flink-pods-9500” Of course, any of these fields may be stored in the same table, related tables, or related databases via joins or associations. The time series database **240** may be combined with the database **210** or maintained separately. The time series database **240** may also collect and store other data from the change stream processing application **230** for performance and error monitoring. Such other data may comprise measurements or events that are tracked, monitored, and aggregated over time, and may include server metrics, application performance monitoring, network data, sensor data, events, clicks, and many other types of analytics data.

(40) Referring back to FIG. 2, in some example embodiments, the application starter **250** is

configured to retrieve an offset value from the time series database **240** using a time parameter. For example, the application starter **250** may synchronously read the offset value from the time series database **240** via hypertext transfer protocol (HTTP) using the time parameter. The time parameter may comprise a single point in time (e.g., a specific date and time of day) or a time range having a start time (e.g., a specific start date and start time) and an end time (e.g., a specific end date and end time). However, other types of time parameters are also within the scope of the present disclosure. In some example embodiments, the application starter **250** issues a query comprising the time parameter to the time series database **240** to retrieve the offset value. Below is one example of a query that may be used to find the offset(s) that belong to a component “x”, located at data center “slc”, and other filters for a given time range:

(41) TABLE-US-00001 `http://<prometheus-server>:<port>/api/v1/query_range?query=monstor_cdc_partitionOffsets{ label1="x",label2="slc",label3="ItemListing|User",label4="NA"} &start=2018-02-04T08:53:34.767Z &end=2018-02-04T08:53:39.854Z &step=20s`

And, below is one example of a corresponding response to the above query with the offset highlighted in bold:

(42) TABLE-US-00002 `monstor_cdc_partitionOffsets{ clusterName="rhs-sample-tm",componentID="3081",datacenter="slc",feeder="gsi",instance="10.000.000.87:950",job="gsi-flink-pods-9500",keyspace="Listing",kubernetes_io_hostname="yyy-node-z21w6-2125230.phx02.dev.ebayc3.com", memberID="1",milkyway="milkyway-staging",name="rhs-sample-tm-phx",namespace="rheos-streaming-qa", offset="250439687",partition="1",pod_name="rhs-sample-tm-phx-1128650763-14jmn", pod_template_hash="1128650755",region="NA",serviced="5253",serviceName="rhs-sample",serviceType="flink",topic="item.t1",type="flink",zone="phx"}`

(43) In some example embodiments, the application starter **250** is configured to transmit a data request comprising the retrieved offset value to the change stream processing application **230**. The data request may be configured or otherwise operable to retrieve a data record stored in association with the retrieved offset value in a storage layer of the stream-processing platform using the offset value. For example, the data request may be configured to cause the change stream processing application **230** to look up and return the data record corresponding to the offset value included in the data request from the log files **222** in the distributed data store **220**. FIG. 4 illustrates a distributed data store **220** for log files **222**, in accordance with some example embodiments.

(44) In the example embodiment of FIG. 4, the data records comprise change data (e.g., CHANGE DATA-1, CHANGE DATA-2, . . . , CHANGE DATA-N) that are each stored in association with their corresponding offset value (e.g., OFFSET-1, OFFSET-2, . . . , OFFSET-N). Each change data may indicate at least one change to content of an online site. In some example embodiments, each change data, or other type of data record, is also stored in association with one or more other attributes, such as an identification of a corresponding datacenter for the data record and an identification of a corresponding topic for the data record. Other attributes are also within the scope of the present disclosure.

(45) In some example embodiments, the data request is further operable to reposition a stream of data published from the streaming platform system **150** to an application at a position corresponding to the offset value included in the data request. The retrieval of the offset value by the application starter **250** and the use of the retrieved offset value in the data request from the application starter **250** to the change stream processing application **230** may be performed in response to, or otherwise based on, an interruption of the change stream processing application **230** publishing a stream of data to an application. For example, the application starter **250** may detect that a network or system failure has occurred and be triggered by such failure detection to retrieve the offset value and use it in a data request to the change stream processing application **230** replay or reposition the processing of the change stream processing application **230** to a position corresponding to a point in time just before the network or system failure.

(46) By storing the offset values of the data records in the time series database **240**, the streaming platform system **150** of the present disclosure solves the technical problems of prior solutions and improves the functioning of the underlying computer system. The storage of the offset values in the time series database **240** makes it straightforward and easy for the application starter **250** to query for the offset value that corresponds to a particular point in time, thereby enabling the processors of the streaming platform system **150** to reprocess the database changes, or other data records, from a specific timestamp or that belong to a time range between start and end timestamps. As an additional feature, tracking the offset values this way provides another way of monitoring the liveliness of the data stream of the streaming platform system **150**, as the offset values can be queried and displayed as a metric. The streaming platform system **150** provides a multipurpose time series database that offers both offset management separate from the storage of the data records in the storage layer of the streaming platform system **150** and use of the time series database **240** to monitor metrics of the change stream processing application **230**.

(47) Additionally, the use of the time series database **240** to store the offset values of data records of the data stream of the streaming platform system **150** improves the accuracy, effectiveness, and efficiency of accessing specific points in the data stream, as well as facilitates recovery of the streaming platform system **150** from network or system failure. This use of the time series database **240** to store the offset values also reduces data query errors, since the application starter **250** may accurately retrieve the most appropriate offset value using a specific time parameter for the specific data record being targeted by the application starter **250**. Additionally, the use of the time series database **240** to manage offsets, as disclosed herein, reduces dependency on a specific version of an API to identify and access a particular point in the data stream, thereby improving the flexibility and accessibility of the streaming platform computer system.

(48) FIG. 5 is a flowchart illustrating a method **500** of improving the functioning of a streaming platform system by managing database change stream offsets using a time series database, in accordance with some example embodiments. The operations of method **500** can be performed by a system or modules of a system. The operations of method **500** can be performed by processing logic that can comprise hardware (e.g., circuitry, dedicated logic, programmable logic, microcode, etc.), software (e.g., instructions run on a processing device), or a combination thereof. In one example embodiment, the method **500** is performed by the streaming platform system **150** of FIGS. 1 and 2, or any combination of one or more of its components or modules (e.g., the application starter **250**, the change stream processing application **230**), as described above.

(49) At operation **510**, the streaming platform system **150** retrieves an offset value from a plurality of offset values stored in a time series database. For example, the application starter **250** may retrieve the offset value from the time series database **240**, as previously discussed. The offset values may be indexed in the time series database **240** in time order and the retrieved offset value may be retrieved using a time parameter. In some example embodiments, the time parameter comprises a single point in time. In some example embodiments, the time parameter comprises a time range having a start time and an end time. Other types of time parameters are also within the scope of the present disclosure. In some example embodiments, the retrieving of the offset value at operation **510** is performed in response to a detection by the application starter **250** of an interruption of the change stream processing application **230** publishing a stream of data to an application. However, other triggers for the performance of the retrieval of the offset value at operation **510** are also within the scope of the present disclosure.

(50) At operation **520**, the streaming platform system **150** transmits a data request to a stream-processing platform. For example, the application starter **250** may transmit the data request to the change stream processing application **230**. In some example embodiments, the data request comprises the retrieved offset value, and the data request is operable to retrieve a data record stored in association with the retrieved offset value in a storage layer of the stream-processing platform using the offset value. For example, the data record may be retrieved from the log file(s) **222** in the

distributed data store **220** using the retrieved offset value as an index to identify the corresponding data record. In some example embodiments, the data record comprises change data indicating at least one change to content of an online site. However, other types of data records are also within the scope of the present disclosure. In some example embodiments, the data request is further operable to reposition a stream of data published from the stream-processing platform to an application at a position corresponding to the retrieved offset value, such as in situations where an interruption of the change stream processing application **230** publishing a stream of data to an application has been detected, as previously discussed.

(51) It is contemplated that the operations of method **500** can incorporate any of the other features disclosed herein.

(52) FIG. **6** is a flowchart illustrating another method **600** of improving the functioning of a streaming platform system by managing database change stream offsets using a time series database, in accordance with some example embodiments. The operations of method **600** can be performed by a system or modules of a system. The operations of method **600** can be performed by processing logic that can comprise hardware (e.g., circuitry, dedicated logic, programmable logic, microcode, etc.), software (e.g., instructions run on a processing device), or a combination thereof. In one example embodiment, the method **600** is performed by the streaming platform system **150** of FIGS. **1** and **2**, or any combination of one or more of its components or modules (e.g., the application starter **250**, the change stream processing application **230**), as described above.

(53) In some example embodiments, the method **600** comprises operations **602**, **604**, and **606** being performed prior to the performance of operations **510** and **520** of the method **500** in FIG. **5**. At operation **602**, the streaming platform system **150** receives a series of data records. For example, the change stream processing application **230** may receive a series of data records comprising change data from the database **210**. At operation **604**, the streaming platform system **150**, for each one of the data records in the series of data records, stores the data record in the storage layer of the stream-processing platform in association with a corresponding one of the plurality of offset values. For example, each data record may be stored in a log file **222** of the distributed data store **220**. At operation **606**, the streaming platform system **150**, for each one of the plurality of offset values, stores the offset value in the time series database. For example, the change stream processing application **230** may store each offset value that is stored in the distributed data store **220** in the time series database **240**. Each offset value may be stored in the time series database **240** in association with a corresponding timestamp for subsequent retrieval using a time parameter.

(54) In some example embodiments, the change stream processing application **230** writes, or otherwise stores, the successfully processed offset values in the time series database **240** at a fixed periodic time. While, in some example embodiments, the change stream processing application **230** may store each successfully processed offset value in the time series database **240**, in other example embodiments, the only offset values that the change stream processing application **240** writes to the time series database **240** are the offset values that are successfully processed at a time corresponding to the fixed periodic time. In one example where the change stream processing application **240** successfully processes an offset value X at time T1, an offset value Y at time T2, and an offset value Z at time T3, the change stream processing application **240** may only store the offset value Z, and not offset values X and Y, in the time series database **240** if T3 corresponds to the fixed periodic time at which the offset values are written to the time series database **240**.

(55) It is contemplated that the operations of method **600** can incorporate any of the other features disclosed herein.

(56) FIG. **7** is a block diagram illustrating various components of the network-based publication system **142**, in accordance with some example embodiments. The publication system **142** can be hosted on dedicated or shared server machines that are communicatively coupled to enable communications between server machines. The components themselves are communicatively coupled (e.g., via appropriate interfaces) to each other and to various data sources, so as to allow

information to be passed between the components or so as to allow the components to share and access common data. Furthermore, the components can access one or more databases **126** via the database servers **124**.

(57) The publication system **142** can provide a number of publishing, listing, and/or price-setting mechanisms whereby a seller (also referred to as a first user) can list (or publish information concerning) goods or services for sale or barter, a buyer (also referred to as a second user) can express interest in or indicate a desire to purchase or barter such goods or services, and a transaction (such as a trade) can be completed pertaining to the goods or services. To this end, the publication system **142** can comprise at least one publication engine **702** and one or more selling engines **704**. The publication engine **702** can publish information, such as item listings or product description pages, on the publication system **142**. In some embodiments, the selling engines **704** can comprise one or more fixed-price engines that support fixed-price listing and price setting mechanisms and one or more auction engines that support auction-format listing and price setting mechanisms (e.g., English, Dutch, Chinese, Double, Reverse auctions, etc.). The various auction engines can also provide a number of features in support of these auction-format listings, such as a reserve price feature whereby a seller can specify a reserve price in connection with a listing and a proxy-bidding feature whereby a bidder can invoke automated proxy bidding. The selling engines **704** can further comprise one or more deal engines that support merchant-generated offers for products and services.

(58) A listing engine **706** allows sellers to conveniently author listings of items or authors to author publications. In one embodiment, the listings pertain to goods or services that a user (e.g., a seller) wishes to transact via the publication system **142**. In some embodiments, the listings can be an offer, deal, coupon, or discount for the good or service. Each good or service is associated with a particular category. The listing engine **706** can receive listing data such as title, description, and aspect name/value pairs. Furthermore, each listing for a good or service can be assigned an item identifier. In other embodiments, a user can create a listing that is an advertisement or other form of information publication. The listing information can then be stored to one or more storage devices coupled to the publication system **142** (e.g., databases **126**). Listings also can comprise product pages that display a product and information (e.g., product title, specifications, descriptions, and reviews) associated with the product. In some embodiments, the product page can include an aggregation of item listings that correspond to the product described on the product page.

(59) The listing engine **706** can also allow buyers to conveniently author listings or requests for items desired to be purchased. In some embodiments, the listings can pertain to goods or services that a user (e.g., a buyer) wishes to transact via the publication system **142**. Each good or service is associated with a particular category. The listing engine **706** can receive as much or as little listing data, such as title, description, and aspect name/value pairs, that the buyer is aware of about the requested item. In some embodiments, the listing engine **706** can parse the buyer's submitted item information and can complete incomplete portions of the listing. For example, if the buyer provides a brief description of a requested item, the listing engine **706** can parse the description, extract key terms, and use those terms to make a determination of the identity of the item. Using the determined item identity, the listing engine **706** can retrieve additional item details for inclusion in the buyer item request. In some embodiments, the listing engine **706** can assign an item identifier to each listing for a good or service.

(60) In some embodiments, the listing engine **706** allows sellers to generate offers for discounts on products or services. The listing engine **706** can receive listing data, such as the product or service being offered, a price and/or discount for the product or service, a time period for which the offer is valid, and so forth. In some embodiments, the listing engine **706** permits sellers to generate offers from the sellers' mobile devices. The generated offers can be uploaded to the publication system **142** for storage and tracking.

(61) Searching the publication system **142** is facilitated by a searching engine **708**. For example,

the searching engine **708** enables keyword queries of listings published via the publication system **142**. In example embodiments, the searching engine **708** receives the keyword queries from a device of a user and conducts a review of the storage device storing the listing information. The review will enable compilation of a result set of listings that can be sorted and returned to the client device **110** of the user. The searching engine **708** can record the query (e.g., keywords) and any subsequent user actions and behaviors (e.g., navigations).

(62) The searching engine **708** also can perform a search based on the location of the user. A user can access the searching engine **708** via a mobile device and generate a search query. Using the search query and the user's location, the searching engine **708** can return relevant search results for products, services, offers, auctions, and so forth to the user. The searching engine **708** can identify relevant search results both in a list form and graphically on a map. Selection of a graphical indicator on the map can provide additional details regarding the selected search result. In some embodiments, the user can specify as part of the search query a radius or distance from the user's current location to limit search results.

(63) The searching engine **708** also can perform a search based on an image. The image can be taken from a camera or imaging component of a client device or can be accessed from storage.

(64) In a further example, a navigation engine **710** allows users to navigate through various categories, catalogs, or inventory data structures according to which listings can be classified within the publication system **142**. For example, the navigation engine **710** allows a user to successively navigate down a category tree comprising a hierarchy of categories (e.g., the category tree structure) until a particular set of listings is reached. Various other navigation applications within the navigation engine **710** can be provided to supplement the searching and browsing applications. The navigation engine **710** can record the various user actions (e.g., clicks) performed by the user in order to navigate down the category tree.

(65) FIG. **8** is a high-level entity-relationship diagram, illustrating various tables **800** that can be maintained within the database(s) **126**, and that are utilized by and support the systems **142**, **144**, and **150**. A user table **802** contains a record for each registered user of the networked system **102**, and can include identifier, address and financial instrument information pertaining to each such registered user. A user can operate as a seller, a buyer, or both, within the networked system **102**. In one example embodiment, a buyer can be a user that has accumulated value (e.g., commercial or proprietary currency), and is accordingly able to exchange the accumulated value for items that are offered for sale by the networked system **102**.

(66) The tables **800** also include an items table **804** in which are maintained item records for goods and services that are available to be, or have been, transacted via the networked system **102**. Each item record within the items table **804** can furthermore be linked to one or more user records within the user table **802**, so as to associate a seller and one or more actual or potential buyers with each item record.

(67) A transaction table **806** contains a record for each transaction (e.g., a purchase or sale transaction) pertaining to items for which records exist within the items table **804**. An order table **808** is populated with order records, with each order record being associated with an order. Each order, in turn, can be associated with one or more transactions for which records exist within the transaction table **806**.

(68) Bid records within a bids table **810** each relate to a bid received at the networked system **102** in connection with an auction-format listing supported by an auction application. A feedback table **812** is utilized by one or more reputation applications, in one example embodiment, to construct and maintain reputation information concerning users. A history table **814** maintains a history of transactions to which a user has been a party. One or more attributes tables **816** record attribute information pertaining to items for which records exist within the items table **804**. Considering only a single example of such an attribute, the attributes tables **816** can indicate a currency attribute associated with a particular item, with the currency attribute identifying the currency of a price for

the relevant item as specified by a seller.

(69) In some example embodiments, the tables **800** also include a products table **818** in which are maintained product records for goods and services that are available to be, or have been, transacted via the networked system **102**. Each product record within the products table **818** can furthermore be linked to one or more user records within the user table **802**, so as to associate a seller and one or more actual or potential buyers with each item record. In some example embodiments, a reviews table **820** is utilized by one or more review applications, in one example embodiment, to construct and maintain reviews concerning products. Such product reviews can be authored and submitted by users and can include, but are not limited to, text-based information describing a user's experience with the corresponding product, as well as one or more user-submitted ratings of the product, such as one or more graphical user interface elements that represent a rating (e.g., a star-based ratings system).

(70) The streaming platform system **150** may be communicatively coupled to (e.g., in communication with) any of the components of the publication system **142** in FIG. 7 or any of the tables **800** in FIG. 8. In some example embodiments, the data records that are processed by the streaming platform system **150** comprise any of combination of the data discussed above with respect to the publication system **142** in FIG. 7 or the tables **800** in FIG. 8. However, other types of data of the data records are also within the scope of the present disclosure.

(71) It is to be appreciated the embodiments described herein will work with any system requiring data to be sourced and stored at a high rate—they may be used in cloud environments, data center environments, on servers, desktops, or mobile devices. It may be, for example, that a system to facilitate storage and subsequent access to high volume data is available as a web service from a cloud data center provider.

(72) It is contemplated that any features of any embodiments disclosed herein can be combined with any other features of any other embodiments disclosed herein. Accordingly, any such hybrid embodiments are within the scope of the present disclosure.

(73) Certain embodiments are described herein as including logic or a number of components, modules, or mechanisms. Modules may constitute either software modules (e.g., code embodied on a machine-readable medium) or hardware modules. A “hardware module” is a tangible unit capable of performing certain operations and may be configured or arranged in a certain physical manner. In various example embodiments, one or more computer systems (e.g., a standalone computer system, a client computer system, or a server computer system) or one or more hardware modules of a computer system (e.g., a processor or a group of processors) may be configured by software (e.g., an application or application portion) as a hardware module that operates to perform certain operations as described herein.

(74) In some embodiments, a hardware module may be implemented mechanically, electronically, or any suitable combination thereof. For example, a hardware module may include dedicated circuitry or logic that is permanently configured to perform certain operations. For example, a hardware module may be a special-purpose processor, such as a Field-Programmable Gate Array (FPGA) or an Application Specific Integrated Circuit (ASIC). A hardware module may also include programmable logic or circuitry that is temporarily configured by software to perform certain operations. For example, a hardware module may include software executed by a programmable processor. Once configured by such software, hardware modules become specific machines (or specific components of a machine) uniquely tailored to perform the configured functions. It will be appreciated that the decision to implement a hardware module mechanically, in dedicated and permanently configured circuitry, or in temporarily configured circuitry (e.g., configured by software) may be driven by cost and time considerations.

(75) Accordingly, the phrase “hardware module” should be understood to encompass a tangible entity, be that an entity that is physically constructed, permanently configured (e.g., hardwired), or temporarily configured (e.g., programmed) to operate in a certain manner or to perform certain

operations described herein. As used herein, “hardware-implemented module” refers to a hardware module. Considering embodiments in which hardware modules are temporarily configured (e.g., programmed), each of the hardware modules need not be configured or instantiated at any one instance in time. For example, where a hardware module comprises a processor configured by software to become a special-purpose processor, the processor may be configured as respectively different special-purpose processors (e.g., comprising different hardware modules) at different times. Software accordingly configures a particular processor or processors, for example, to constitute a particular hardware module at one instance of time and to constitute a different hardware module at a different instance of time.

(76) Hardware modules can provide information to, and receive information from, other hardware modules. Accordingly, the described hardware modules may be regarded as being communicatively coupled. Where multiple hardware modules exist contemporaneously, communications may be achieved through signal transmission (e.g., over appropriate circuits and buses) between or among two or more of the hardware modules. In embodiments in which multiple hardware modules are configured or instantiated at different times, communications between such hardware modules may be achieved, for example, through the storage and retrieval of information in memory structures to which the multiple hardware modules have access. For example, one hardware module may perform an operation and store the output of that operation in a memory device to which it is communicatively coupled. A further hardware module may then, at a later time, access the memory device to retrieve and process the stored output. Hardware modules may also initiate communications with input or output devices, and can operate on a resource (e.g., a collection of information).

(77) The various operations of example methods described herein may be performed, at least partially, by one or more processors that are temporarily configured (e.g., by software) or permanently configured to perform the relevant operations. Whether temporarily or permanently configured, such processors may constitute processor-implemented modules that operate to perform one or more operations or functions described herein. As used herein, “processor-implemented module” refers to a hardware module implemented using one or more processors.

(78) Similarly, the methods described herein may be at least partially processor-implemented, with a particular processor or processors being an example of hardware. For example, at least some of the operations of a method may be performed by one or more processors or processor-implemented modules. Moreover, the one or more processors may also operate to support performance of the relevant operations in a “cloud computing” environment or as a “software as a service” (SaaS). For example, at least some of the operations may be performed by a group of computers (as examples of machines including processors), with these operations being accessible via a network (e.g., the Internet) and via one or more appropriate interfaces (e.g., an Application Program Interface (API)).

(79) The performance of certain of the operations may be distributed among the processors, not only residing within a single machine, but deployed across a number of machines. In some example embodiments, the processors or processor-implemented modules may be located in a single geographic location (e.g., within a home environment, an office environment, or a server farm). In other example embodiments, the processors or processor-implemented modules may be distributed across a number of geographic locations.

(80) The modules, methods, applications and so forth described in conjunction with FIGS. 1-8 are implemented in some embodiments in the context of a machine and an associated software architecture. The sections below describe representative software architecture(s) and machine (e.g., hardware) architecture that are suitable for use with the disclosed embodiments.

(81) Software architectures are used in conjunction with hardware architectures to create devices and machines tailored to particular purposes. For example, a particular hardware architecture coupled with a particular software architecture will create a mobile device, such as a mobile phone, tablet device, or so forth. A slightly different hardware and software architecture may yield a smart

device for use in the “internet of things.” While yet another combination produces a server computer for use within a cloud computing architecture. Not all combinations of such software and hardware architectures are presented here as those of skill in the art can readily understand how to implement the features of the present disclosure in different contexts from the disclosure contained herein.

(82) FIG. 9 is a block diagram **900** illustrating a representative software architecture **902**, which may be used in conjunction with various hardware architectures herein described. FIG. 9 is merely a non-limiting example of a software architecture **902** and it will be appreciated that many other architectures may be implemented to facilitate the functionality described herein. The software architecture **902** may be executing on hardware such as a machine **1000** of FIG. 10 that includes, among other things, processors **1010**, memory/storage **1030**, and I/O components **1050**. A representative hardware layer **904** is illustrated in FIG. 9 and can represent, for example, the machine **1000** of FIG. 10. The representative hardware layer **904** comprises one or more processing units **906** having associated executable instructions **908**. The executable instructions **908** represent the executable instructions of the software architecture **902**, including implementation of the methods, modules, and features disclosed above with respect to FIGS. 1-8. The hardware layer **904** also includes memory and/or storage modules **910**, which also have the executable instructions **908**. The hardware layer **904** may also comprise other hardware **912**, which represents any other hardware of the hardware layer **904**, such as the other hardware illustrated as part of the machine **1000**.

(83) In the example architecture of FIG. 9, the software architecture **902** may be conceptualized as a stack of layers where each layer provides particular functionality. For example, the software architecture **902** may include layers such as an operating system **914**, libraries **916**, frameworks/middleware **918**, applications **920**, and a presentation layer **944**. Operationally, the applications **920** and/or other components within the layers may invoke application programming interface (API) calls **924** through the software stack and receive a response, returned values, and so forth, illustrated as messages **926**, in response to the API calls **924**. The layers illustrated are representative in nature and not all software architectures have all layers. For example, some mobile or special purpose operating systems may not provide a frameworks/middleware **918**, while others may provide such a layer. Other software architectures may include additional or different layers.

(84) The operating system **914** may manage hardware resources and provide common services. The operating system **914** may include, for example, a kernel **928**, services **930**, and drivers **932**. The kernel **928** may act as an abstraction layer between the hardware and the other software layers. For example, the kernel **928** may be responsible for memory management, processor management (e.g., scheduling), component management, networking, security settings, and so on. The services **930** may provide other common services for the other software layers. The drivers **932** may be responsible for controlling or interfacing with the underlying hardware. For instance, the drivers **932** may include display drivers, camera drivers, Bluetooth® drivers, flash memory drivers, serial communication drivers (e.g., Universal Serial Bus (USB) drivers), Wi-Fi® drivers, audio drivers, power management drivers, and so forth depending on the hardware configuration.

(85) The libraries **916** may provide a common infrastructure that may be utilized by the applications **920** or other components or layers. The libraries **916** typically provide functionality that allows other software modules to perform tasks in an easier fashion than to interface directly with the underlying operating system **914** functionality (e.g., kernel **928**, services **930**, and/or drivers **932**). The libraries **916** may include system libraries **934** (e.g., C standard library) that may provide functions such as memory allocation functions, string manipulation functions, mathematic functions, and the like. In addition, the libraries **916** may include API libraries **936** such as media libraries (e.g., libraries to support presentation and manipulation of various media formats such as MPEG4, H.264, MP3, AAC, AMR, JPG, PNG), graphics libraries (e.g., an OpenGL framework

that may be used to render 2D and 3D graphic content on a display), database libraries (e.g., SQLite that may provide various relational database functions), web libraries (e.g., WebKit that may provide web browsing functionality), and the like. The libraries **916** may also include a wide variety of other libraries **938** to provide many other APIs to the applications **920** and other software components/modules.

(86) The frameworks/middleware **918** may provide a higher-level common infrastructure that may be utilized by the applications **920** or other software components/modules. For example, the frameworks/middleware **918** may provide various graphic user interface (GUI) functions, high-level resource management, high-level location services, and so forth. The frameworks/middleware **918** may provide a broad spectrum of other APIs that may be utilized by the applications **920** or other software components/modules, some of which may be specific to a particular operating system or platform.

(87) The applications **920** include built-in applications **940** or third party applications **942**. Examples of representative built-in applications **940** may include, but are not limited to, a contacts application, a browser application, a book reader application, a location application, a media application, a messaging application, or a game application. The third party applications **942** may include any of the built in applications **940** as well as a broad assortment of other applications. In a specific example, the third party application **942** (e.g., an application developed using the Android™ or iOS™ software development kit (SDK) by an entity other than the vendor of the particular platform) may be mobile software running on a mobile operating system such as iOS™, Android™, Windows® Phone, or other mobile operating systems. In this example, the third party application **942** may invoke the API calls **924** provided by the mobile operating system such as the operating system **914** to facilitate functionality described herein.

(88) The applications **920** may utilize built-in operating system functions (e.g., kernel **928**, services **930**, and/or drivers **932**), libraries (e.g., system libraries **934**, API libraries **936**, and other libraries **938**), and frameworks/middleware **918** to create user interfaces to interact with users of the system. Alternatively, or additionally, in some systems, interactions with a user may occur through a presentation layer, such as the presentation layer **944**. In these systems, the application/module “logic” can be separated from the aspects of the application/module that interact with a user.

(89) Some software architectures utilize virtual machines. In the example of FIG. 9, this is illustrated by a virtual machine **948**. A virtual machine creates a software environment where applications/modules can execute as if they were executing on a hardware machine (e.g., the machine of FIG. 10). A virtual machine is hosted by a host operating system (e.g., operating system **914**) and typically, although not always, has a virtual machine monitor **946**, which manages the operation of the virtual machine **948** as well as the interface with the host operating system (e.g., operating system **914**). A software architecture executes within the virtual machine **948** such as an operating system **950**, libraries **952**, frameworks **954**, applications **956**, or presentation layer **958**. These layers of software architecture executing within the virtual machine **948** can be the same as corresponding layers previously described or may be different.

(90) FIG. 10 is a block diagram illustrating components of a machine **1000**, according to some example embodiments, able to read instructions from a machine-readable medium (e.g., a machine-readable storage medium) and perform any one or more of the methodologies discussed herein. Specifically, FIG. 10 shows a diagrammatic representation of the machine **1000** in the example form of a computer system, within which instructions **1016** (e.g., software, a program, an application, an applet, an app, or other executable code) for causing the machine **1000** to perform any one or more of the methodologies discussed herein may be executed. For example, the instructions may cause the machine **1000** to execute any one of the respective methods **500** and **600** of FIGS. 5 and 6. Additionally, or alternatively, the instructions **1016** may implement any combination of one or more of the components of the streaming platform system **150** of FIG. 2, and so forth. The instructions **1016** transform the non-programmed machine into a particular machine

programmed to carry out the described and illustrated functions in the manner described. In alternative embodiments, the machine **1000** operates as a standalone device or may be coupled (e.g., networked) to other machines. In a networked deployment, the machine **1000** may operate in the capacity of a server machine or a client machine in a server-client network environment, or as a peer machine in a peer-to-peer (or distributed) network environment. The machine **1000** may comprise, but not be limited to, a server computer, a client computer, a personal computer (PC), a tablet computer, a laptop computer, a netbook, a set-top box (STB), a personal digital assistant (PDA), an entertainment media system, a cellular telephone, a smart phone, a mobile device, a wearable device (e.g., a smart watch), a smart home device (e.g., a smart appliance), other smart devices, a web appliance, a network router, a network switch, a network bridge, or any machine capable of executing the instructions **1016**, sequentially or otherwise, that specify actions to be taken by machine **1000**. Further, while only a single machine **1000** is illustrated, the term “machine” shall also be taken to include a collection of machines **1000** that individually or jointly execute the instructions **1016** to perform any one or more of the methodologies discussed herein.

(91) The machine **1000** may include processors **1010**, memory **1030**, and I/O components **1050**, which may be configured to communicate with each other such as via a bus **1002**. In an example embodiment, the processors **1010** (e.g., a Central Processing Unit (CPU), a Reduced Instruction Set Computing (RISC) processor, a Complex Instruction Set Computing (CISC) processor, a Graphics Processing Unit (GPU), a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Radio-Frequency Integrated Circuit (RFIC), another processor, or any suitable combination thereof) may include, for example, processor **1012** and processor **1014** that may execute instructions **1016**. The term “processor” is intended to include multi-core processor that may comprise two or more independent processors (sometimes referred to as “cores”) that may execute instructions contemporaneously. Although FIG. **10** shows multiple processors **1010**, the machine **1000** may include a single processor with a single core, a single processor with multiple cores (e.g., a multi-core process), multiple processors with a single core, multiple processors with multiples cores, or any combination thereof.

(92) The memory/storage **1030** may include a memory **1032**, such as a main memory, or other memory storage, and a storage unit **1036**, both accessible to the processors **1010** such as via the bus **1002**. The storage unit **1036** and memory **1032** store the instructions **1016** embodying any one or more of the methodologies or functions described herein. The instructions **1016** may also reside, completely or partially, within the memory **1032**, within the storage unit **1036**, within at least one of the processors **1010** (e.g., within the processor's cache memory), or any suitable combination thereof, during execution thereof by the machine **1000**. Accordingly, the memory **1032**, the storage unit **1036**, and the memory of processors **1010** are examples of machine-readable media.

(93) As used herein, “machine-readable medium” means a device able to store or carry instructions and data temporarily or permanently and may include, but is not be limited to, random-access memory (RAM), read-only memory (ROM), buffer memory, flash memory, optical media, magnetic media, cache memory, other types of storage (e.g., Erasable Programmable Read-Only Memory (EEPROM)) and/or any suitable combination thereof. The term “machine-readable medium” should be taken to include a single medium or multiple media (e.g., a centralized or distributed database, or associated caches and servers) able to store instructions **1016**. The term “machine-readable medium” shall also be taken to include any medium, or combination of multiple media, that is capable of storing or carrying instructions (e.g., instructions **1016**) for execution by a machine (e.g., machine **1000**), such that the instructions, when executed by one or more processors of the machine **1000** (e.g., processors **1010**), cause the machine **1000** to perform any one or more of the methodologies described herein. Accordingly, a “machine-readable storage medium” refers to a single storage apparatus or device, as well as “cloud-based” storage systems or storage networks that include multiple storage apparatus or devices. The term “machine-readable medium” includes transmission media such as signals.

(94) The I/O components **1050** may include a wide variety of components to receive input, provide output, produce output, transmit information, exchange information, capture measurements, and so on. The specific I/O components **1050** that are included in a particular machine will depend on the type of machine. For example, portable machines such as mobile phones will likely include a touch input device or other such input mechanisms, while a headless server machine will likely not include such a touch input device. It will be appreciated that the I/O components **1050** may include many other components that are not shown in FIG. **10**. The I/O components **1050** are grouped according to functionality merely for simplifying the following discussion and the grouping is in no way limiting. In various example embodiments, the I/O components **1050** may include output components **1052** and input components **1054**. The output components **1052** may include visual components (e.g., a display such as a plasma display panel (PDP), a light emitting diode (LED) display, a liquid crystal display (LCD), a projector, or a cathode ray tube (CRT)), acoustic components (e.g., speakers), haptic components (e.g., a vibratory motor, resistance mechanisms), other signal generators, and so forth. The input components **1054** may include alphanumeric input components (e.g., a keyboard, a touch screen configured to receive alphanumeric input, a photo-optical keyboard, or other alphanumeric input components), point based input components (e.g., a mouse, a touchpad, a trackball, a joystick, a motion sensor, or other pointing instrument), tactile input components (e.g., a physical button, a touch screen that provides location and/or force of touches or touch gestures, or other tactile input components), audio input components (e.g., a microphone), and the like.

(95) In further example embodiments, the I/O components **1050** may include biometric components **1056**, motion components **1058**, environmental components **1060**, or position components **1062** among a wide array of other components. For example, the biometric components **1056** may include components to detect expressions (e.g., hand expressions, facial expressions, vocal expressions, body gestures, or eye tracking), measure biosignals (e.g., blood pressure, heart rate, body temperature, perspiration, or brain waves), identify a person (e.g., voice identification, retinal identification, facial identification, fingerprint identification, or electroencephalogram based identification), and the like. The motion components **1058** may include acceleration sensor components (e.g., accelerometer), gravitation sensor components, rotation sensor components (e.g., gyroscope), and so forth. The environmental components **1060** may include, for example, illumination sensor components (e.g., photometer), temperature sensor components (e.g., one or more thermometer that detect ambient temperature), humidity sensor components, pressure sensor components (e.g., barometer), acoustic sensor components (e.g., one or more microphones that detect background noise), proximity sensor components (e.g., infrared sensors that detect nearby objects), gas sensors (e.g., gas detection sensors to detection concentrations of hazardous gases for safety or to measure pollutants in the atmosphere), or other components that may provide indications, measurements, or signals corresponding to a surrounding physical environment. The position components **1062** may include location sensor components (e.g., a Global Position System (GPS) receiver component), altitude sensor components (e.g., altimeters or barometers that detect air pressure from which altitude may be derived), orientation sensor components (e.g., magnetometers), and the like.

(96) Communication may be implemented using a wide variety of technologies. The I/O components **1050** may include communication components **1064** operable to couple the machine **1000** to a network **1080** or devices **1070** via coupling **1082** and coupling **1072** respectively. For example, the communication components **1064** may include a network interface component or other suitable device to interface with the network **1080**. In further examples, communication components **1064** may include wired communication components, wireless communication components, cellular communication components, Near Field Communication (NFC) components, Bluetooth® components (e.g., Bluetooth® Low Energy), Wi-Fi® components, and other communication components to provide communication via other modalities. The devices **1070** may

be another machine or any of a wide variety of peripheral devices (e.g., a peripheral device coupled via a Universal Serial Bus (USB)).

(97) Moreover, the communication components **1064** may detect identifiers or include components operable to detect identifiers. For example, the communication components **1064** may include Radio Frequency Identification (RFID) tag reader components, NFC smart tag detection components, optical reader components (e.g., an optical sensor to detect one-dimensional bar codes such as Universal Product Code (UPC) bar code, multi-dimensional bar codes such as Quick Response (QR) code, Aztec code, Data Matrix, Dataglyph, MaxiCode, PDF417, Ultra Code, UCC RSS-2D bar code, and other optical codes), or acoustic detection components (e.g., microphones to identify tagged audio signals). In addition, a variety of information may be derived via the communication components **1064**, such as, location via Internet Protocol (IP) geo-location, location via Wi-Fi® signal triangulation, location via detecting a NFC beacon signal that may indicate a particular location, and so forth.

(98) In various example embodiments, one or more portions of the network **1080** may be an ad hoc network, an intranet, an extranet, a virtual private network (VPN), a local area network (LAN), a wireless LAN (WLAN), a wide area network (WAN), a wireless WAN (WWAN), a metropolitan area network (MAN), the Internet, a portion of the Internet, a portion of the Public Switched Telephone Network (PSTN), a plain old telephone service (POTS) network, a cellular telephone network, a wireless network, a Wi-Fi® network, another type of network, or a combination of two or more such networks. For example, the network **1080** or a portion of the network **1080** may include a wireless or cellular network and the coupling **1082** may be a Code Division Multiple Access (CDMA) connection, a Global System for Mobile communications (GSM) connection, or other type of cellular or wireless coupling. In this example, the coupling **1082** may implement any of a variety of types of data transfer technology, such as Single Carrier Radio Transmission Technology (1×RTT), Evolution-Data Optimized (EVDO) technology, General Packet Radio Service (GPRS) technology, Enhanced Data rates for GSM Evolution (EDGE) technology, third Generation Partnership Project (3GPP) including 3G, fourth generation wireless (4G) networks, Universal Mobile Telecommunications System (UMTS), High Speed Packet Access (HSPA), Worldwide Interoperability for Microwave Access (WiMAX), Long Term Evolution (LTE) standard, others defined by various standard setting organizations, other long range protocols, or other data transfer technology.

(99) The instructions **1016** may be transmitted or received over the network **1080** using a transmission medium via a network interface device (e.g., a network interface component included in the communication components **1064**) and utilizing any one of a number of well-known transfer protocols (e.g., hypertext transfer protocol (HTTP)). Similarly, the instructions **1016** may be transmitted or received using a transmission medium via the coupling **1072** (e.g., a peer-to-peer coupling) to devices **1070**. The term “transmission medium” shall be taken to include any computer-readable medium that is capable of encoding or carrying instructions **1016** for execution by the machine **1000**, and includes digital or analog communications signals or other intangible medium to facilitate communication of such software.

(100) In some example embodiments, any combination of one or more of the components of the software architecture **902** of FIG. 9, the representative hardware layer **904** of FIG. 9, and the machine **1000** of FIG. 10 are incorporated into the streaming platform system **150** to provide the software or hardware architecture of the streaming platform system **150**.

(101) The following numbered examples are embodiments. 1. A computer-implemented method comprising: retrieving, by one or more hardware processors, an offset value from a plurality of offset values stored in a time series database, the plurality of offset values being indexed in the time series database in time order, and the retrieved offset value being retrieved using a time parameter; and transmitting, by the one or more hardware processors, a data request to a stream-processing platform, the data request comprising the retrieved offset value, and the data request being operable

to retrieve a data record stored in association with the retrieved offset value in a storage layer of the stream-processing platform using the offset value. 2. The computer-implemented method of example 1, further comprising: prior to the retrieving of the offset value, receiving, by the one or more hardware processors, a series of data records; for each one of the data records in the series of data records, storing, by the one or more hardware processors, the one of the data records in the storage layer of the stream-processing platform in association with a corresponding one of the plurality of offset values; and for each one of the plurality of offset values, storing, by the one or more hardware processors, the one of the plurality of offset values in the time series database. 3. The computer-implemented method of example 1 or example 2, wherein the data request is further operable to reposition a stream of data published from the stream-processing platform to an application at a position corresponding to the offset value. 4. The computer-implemented method of any one of examples 1 to 3, wherein the time parameter comprises a single point in time. 5. The computer-implemented method of any one of examples 1 to 4, wherein the time parameter comprises a time range having a start time and an end time. 6. The computer-implemented method of any one of examples 1 to 5, wherein the data record comprises change data indicating at least one change to content of an online site. 7. The computer-implemented method of any one of examples 1 to 6, wherein the retrieving of the offset value is performed in response to an interruption of the stream-processing platform publishing a stream of data to an application. 8. A system comprising: at least one processor; and a non-transitory computer-readable medium storing executable instructions that, when executed, cause the at least one processor to perform the method of any one of examples 1 to 7. 10. A non-transitory machine-readable storage medium, tangibly embodying a set of instructions that, when executed by at least one processor, causes the at least one processor to perform the method of any one of examples 1 to 7. 11. A machine-readable medium carrying a set of instructions that, when executed by at least one processor, causes the at least one processor to carry out the method of any one of examples 1 to 7.

(102) Throughout this specification, plural instances may implement components, operations, or structures described as a single instance. Although individual operations of one or more methods are illustrated and described as separate operations, one or more of the individual operations may be performed concurrently, and nothing requires that the operations be performed in the order illustrated. Structures and functionality presented as separate components in example configurations may be implemented as a combined structure or component. Similarly, structures and functionality presented as a single component may be implemented as separate components. These and other variations, modifications, additions, and improvements fall within the scope of the subject matter herein.

(103) Although an overview of the inventive subject matter has been described with reference to specific example embodiments, various modifications and changes may be made to these embodiments without departing from the broader scope of embodiments of the present disclosure. Such embodiments of the inventive subject matter may be referred to herein, individually or collectively, by the term “invention” merely for convenience and without intending to voluntarily limit the scope of this application to any single disclosure or inventive concept if more than one is, in fact, disclosed.

(104) The embodiments illustrated herein are described in sufficient detail to enable those skilled in the art to practice the teachings disclosed. Other embodiments may be used and derived therefrom, such that structural and logical substitutions and changes may be made without departing from the scope of this disclosure. The Detailed Description, therefore, is not to be taken in a limiting sense, and the scope of various embodiments is defined only by the appended claims, along with the full range of equivalents to which such claims are entitled.

(105) As used herein, the term “or” may be construed in either an inclusive or exclusive sense. Moreover, plural instances may be provided for resources, operations, or structures described herein as a single instance. Additionally, boundaries between various resources, operations,

modules, engines, and data stores are somewhat arbitrary, and particular operations are illustrated in a context of specific illustrative configurations. Other allocations of functionality are envisioned and may fall within a scope of various embodiments of the present disclosure. In general, structures and functionality presented as separate resources in the example configurations may be implemented as a combined structure or resource. Similarly, structures and functionality presented as a single resource may be implemented as separate resources. These and other variations, modifications, additions, and improvements fall within a scope of embodiments of the present disclosure as represented by the appended claims. The specification and drawings are, accordingly, to be regarded in an illustrative rather than a restrictive sense.

(106) Although an embodiment has been described with reference to specific example embodiments, it will be evident that various modifications and changes can be made to these embodiments without departing from the broader spirit and scope of the present disclosure. Accordingly, the specification and drawings are to be regarded in an illustrative rather than a restrictive sense. The accompanying drawings that form a part hereof show, by way of illustration, and not of limitation, specific embodiments in which the subject matter can be practiced. The embodiments illustrated are described in sufficient detail to enable those skilled in the art to practice the teachings disclosed herein. Other embodiments can be utilized and derived therefrom, such that structural and logical substitutions and changes can be made without departing from the scope of this disclosure. This Detailed Description, therefore, is not to be taken in a limiting sense, and the scope of various embodiments is defined only by the appended claims, along with the full range of equivalents to which such claims are entitled.

(107) Such embodiments of the inventive subject matter can be referred to herein, individually and/or collectively, by the term “invention” merely for convenience and without intending to voluntarily limit the scope of this application to any single invention or inventive concept if more than one is in fact disclosed. Thus, although specific embodiments have been illustrated and described herein, it should be appreciated that any arrangement calculated to achieve the same purpose can be substituted for the specific embodiments shown. This disclosure is intended to cover any and all adaptations or variations of various embodiments. Combinations of the above embodiments, and other embodiments not specifically described herein, will be apparent to those of skill in the art upon reviewing the above description.

(108) The Abstract of the Disclosure is provided to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in a single embodiment for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed embodiments require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter lies in less than all features of a single disclosed embodiment. Thus the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separate embodiment.

Claims

1. A method implemented by a computing device, the method comprising: receiving a series of data records; positioning processing of the series of data records at a position corresponding to an initial offset value of a plurality of offset values; detecting an interruption in the series of data records; for each one of the data records of the series of data records, storing the one of the data records in a storage layer of a stream-processing platform in association with a corresponding one of the plurality of offset values; querying one or more offset values from the plurality of offset values stored in a time series database by using a time parameter, in which the time parameter corresponds

to a particular time and the one or more offset values correspond to the particular time; repositioning the processing of the series of data records at a position corresponding to the one or more queried offset values, such that the processing resumes from a point in time corresponding to just before the interruption detected; and outputting the one or more offset values for display via a user interface.

2. The method of claim 1, wherein the series of data records correspond to a data stream.
3. The method of claim 1, wherein the particular time is a single point in time.
4. The method of claim 1, wherein the particular time is a time range including a start time and an end time.
5. The method of claim 1, wherein the series of data records comprise change data indicating at least one change to content of an online site.
6. The method of claim 1, wherein the querying is performed in response to an interruption of the stream-processing platform publishing a stream of data to an application.
7. The method of claim 1, further comprising for each one of the plurality of offset values, storing the one of the plurality of offset values in the time series database.
8. The method of claim 1, wherein the plurality of offset values is indexed in the time series database in time order.
9. A system comprising: at least one hardware processor; and a non-transitory memory device storing executable instructions that, when executed, cause the at least one hardware processor to perform operations comprising: receiving a series of data records; positioning processing of the series of data records at a position corresponding to an initial offset value of a plurality of offset values; detecting an interruption in the series of data records; for each one of the data records of the series of data records, storing the one of the data records in a storage layer of a stream-processing platform in association with a corresponding one of the plurality of offset values; querying one or more offset values from the plurality of offset values stored in a time series database by using a time parameter, in which the time parameter corresponds to a particular time and the one or more offset values correspond to the particular time; repositioning the processing of the series of data records at a position corresponding to the one or more queried offset values, such that the processing resumes from a point in time corresponding to just before the interruption detected; and outputting the one or more offset values for display via a user interface.
10. The system of claim 9, wherein the series of data records correspond to a data stream.
11. The system of claim 9, wherein the particular time is a single point in time.
12. The system of claim 9, wherein the particular time is a time range including a start time and an end time.
13. The system of claim 9, wherein the series of data records comprise change data indicating at least one change to content of an online site.
14. The system of claim 9, wherein the querying is performed in response to an interruption of the stream-processing platform publishing a stream of data to an application.
15. The system of claim 9, further comprising for each one of the plurality of offset values, storing the one of the plurality of offset values in the time series database.
16. A non-transitory memory device storing a set of instructions that, when executed by at least one processor, causes the at least one processor to perform operations comprising: receiving a series of data records; positioning processing of the series of data records at a position corresponding to an initial offset value of a plurality of offset values; detecting an interruption in the series of data records; for each one of the data records of the series of data records, storing the one of the data records in a storage layer of a stream-processing platform in association with a corresponding one of the plurality of offset values; querying one or more offset values from the plurality of offset values stored in a time series database by using a time parameter, in which the time parameter corresponds to a particular time and the one or more offset values correspond to the particular time; repositioning the processing of the series of data records at a position corresponding to the one or

more queried offset values, such that the processing resumes from a point in time corresponding to just before the interruption detected; and outputting the one or more offset values for display via a user interface.

17. The non-transitory memory device of claim 16, wherein the series of data records include change data.

18. The non-transitory memory device of claim 16, wherein the series of data records correspond to a data stream.

19. The non-transitory memory device of claim 16, further comprising detecting a triggering condition, and wherein the querying of the one or more offset values is responsive to the detecting of the triggering condition.

20. The non-transitory memory device of claim 16, wherein the querying is performed in response to an interruption of the stream-processing platform publishing a stream of data to an application.
